Autoinflammation With Episodic Fever and Immune Dysregulation (AIFID): SHARPIN Deficiency

Disease: Autoinflammation With Episodic Fever and Immune Dysregulation (AIFID) · **MONDO:** 0968982 · **OMIM:** 620795 · **Causal gene: SHARPIN** (HGNC:25321) · **Category:** Mendelian, autosomal recessive

Summary

Autoinflammation With Episodic Fever and Immune Dysregulation (AIFID), also called SHARPIN deficiency or "sharpenia," is an ultra-rare autosomal-recessive inborn error of immunity caused by biallelic loss-of-function variants in SHARPIN, the gene encoding the accessory/regulatory subunit of the Linear Ubiquitin chain Assembly Complex (LUBAC). The disease was defined in 2024 by Oda and colleagues, who reported two unrelated individuals with homozygous frameshift SHARPIN variants and undetectable SHARPIN protein (PMID: 38609546). SHARPIN is one of three LUBAC subunits (together with the catalytic subunit HOIP/RNF31 and the accessory subunit HOIL-1L/RBCK1). LUBAC is the sole cellular machine that builds Met1 (M1)-linked "linear" polyubiquitin chains, which are deposited on NEMO (IKBKG) and RIPK1 to activate canonical NF-κB signaling and to restrain TNF-superfamily-driven cell death.

Mechanistically, loss of SHARPIN destabilizes LUBAC and reduces M1-linear ubiquitination, which **attenuates canonical NF-κB responses** and **shifts cells from an NF-κB-driven survival program toward TNF-superfamily-mediated apoptosis and necroptosis**. The clinical consequence is a paradoxical combination of **autoinflammation** (recurrent/episodic fever, sterile arthritis, parotitis, colitis, hepatic glycogenosis) and **immunodeficiency** (reduced germinal-center B-cell development due to increased B-cell death). Strikingly, and unlike the Sharpin-null *cpdm* mouse that has long served as the disease model, affected humans have **no dermatitis** and their monocytes show **no cytokine hyperresponsiveness**. The

autoinflammatory component responds dramatically to **TNF inhibition** — one patient achieved complete clinical and transcriptomic resolution on anti-TNF therapy (etanercept) and has remained symptom-free for roughly seven years.

This report synthesizes nine confirmed findings drawn from 24 reviewed papers. The strongest human evidence rests on a single landmark study of two patients, supported by an extensive body of mouse-genetics and biochemical work on SHARPIN/LUBAC spanning more than a decade (three foundational 2011 *Nature*/*Nature Immunology* papers establishing SHARPIN as the third LUBAC subunit, plus subsequent genetic-epistasis studies dissecting the death pathways). AIFID sits within a tightly related family of monogenic "M1-ubiquitinopathies" — HOIP deficiency, HOIL-1 deficiency, and OTULIN-related autoinflammatory syndrome (ORAS) — against which it must be differentially diagnosed.

Section 1 — Disease Information

Overview. AIFID (SHARPIN deficiency) is a Mendelian autoinflammatory disease with concomitant immunodeficiency, caused by complete loss of the SHARPIN protein. The autoinflammation manifests as episodic fever and sterile inflammation of joints, salivary glands, and gut; the immunodeficiency manifests as impaired germinal-center B-cell development and humoral deficiency. It is an inborn error of immunity (included in the IUIS 2024 classification).

Key identifiers.

Resource	Identifier
MONDO	MONDO:0968982
OMIM (phenotype)	620795
OMIM (gene SHARPIN)	611885
HGNC	HGNC:25321
NCBI Gene (Entrez)	81858
UniProt	Q9H0F6
Ensembl	ENSG00000179526
Cytogenetic locus	8q24.3

Synonyms / alternative names. SHARPIN deficiency; AIFID; "sharpenia" (informal, by analogy to otulipenia/HOIL-1 deficiency nomenclature).

Information source. Disease-level characterization is derived from **individual patient reports** (two probands with detailed clinical, immunophenotyping, cellular, and transcriptomic workup) aggregated against mechanistic data from aggregated disease-level and model-organism resources. This is not an EHR-derived or population-registry disease; it is defined case-by-case.

Section 2 — Etiology

Primary cause (genetic). AIFID is caused by **biallelic (homozygous) loss-of-function variants in SHARPIN**. The two reported patients carried homozygous frameshift alleles producing undetectable protein (complete loss of function). There is no environmental or infectious cause of the disease itself, though inflammatory flares are a dysregulated innate-immune response in which TNF-superfamily signaling is the key effector.

Genetic risk factors. The disease is monogenic and fully determined by biallelic SHARPIN LoF. **Consanguinity** is a strong contributing factor: Patient 1 was born to consanguineous parents of Indian origin, consistent with autosomal-recessive inheritance of a rare allele. No susceptibility loci or common-variant modifiers have been defined (sample size of two precludes such analysis).

Environmental risk factors. None established as causal. As with other TNF-driven autoinflammatory diseases, infection or other inflammatory triggers plausibly precipitate flares, but this is inferred rather than demonstrated for AIFID specifically.

Protective factors. No genetic or environmental protective factors are documented for this ultra-rare disease. Mechanistically, anything reducing TNF-superfamily signaling (e.g., anti-TNF therapy) is therapeutically protective against the autoinflammatory phenotype.

Gene–environment interactions. Not characterized in humans. In the mouse model, dsRNA present in skin drives TLR3-mediated keratinocyte death contributing to dermatitis (PMID: 27810922) — a gene (Sharpin-loss) × environment (dsRNA/TLR3 ligand) interaction, though this branch appears not to operate in humans (no dermatitis).

Section 3 — Phenotypes

Phenotypes are compiled from the two reported patients (PMID: 38609546). Frequencies are necessarily qualitative given n=2.

Phenotype	Type	HPO suggestion	Onset	Course	Notes / frequency
Recurrent/ episodic fever	Symptom	HP:0001954 (Recurrent fever)	Childhood	Episodic	Core feature (disease-naming)
Sterile arthritis (ankle, atlanto-axial)	Clinical sign	HP:0001369 (Arthritis)	Childhood	Episodic, progressive disability	P1; sterile synovial fluid; wheelchair-dependent before Rx
Parotitis / parotid swelling	Clinical sign	HP:0040286 (Parotitis)	Age 5 (P1)	Recurrent	P1
Colitis / enteropathy	Clinical sign	HP:0002583 (Colitis)	Childhood	Chronic/ relapsing	P1
Hepatic glycogenosis (liver)	Lab/ pathology	HP:0006568 (Abnormal glycogen storage, liver)	Childhood	—	P1 biopsy suggestive of glycogenosis
Germinal-center B-cell deficiency	Lab abnormality	HP:0010976 (B-cell abnormality)	Childhood	—	Hypoplastic adenoidal GCs; humoral immunodeficiency
Attenuated NF-κB signaling (cellular)	Lab abnormality	—	Congenital (cellular)	—	Fibroblasts & B cells
Febrile episodes with fatal course	Clinical course	—	Childhood	Severe (P2)	P2 died despite anti-TNF

Phenotype characteristics (overall). Age of onset: childhood (parotid swelling from age 5 in P1). Severity: variable — ranging from controllable, anti-TNF-responsive disease (P1) to fatal (P2). Progression: episodic autoinflammation with progressive disability if untreated (P1 was wheelchair-dependent pre-treatment).

Quality-of-life impact. Substantial before treatment: P1 was wheelchair-dependent due to arthritis. Anti-TNF therapy restored function and produced sustained remission (~7 years symptom-free). No formal EQ-5D/SF-36/PROMIS data exist for this ultra-rare disease.

Notable absences (phenotypic discriminators). No dermatitis and **no monocyte cytokine hyperresponsiveness** — both distinguish human SHARPIN deficiency from the *cpdm* mouse and from HOIL-1/HOIP deficiency, respectively.

Section 4 — Genetic / Molecular Information

Causal gene. *SHARPIN* (SHANK-associated RH-domain-interacting protein), HGNC:25321, gene OMIM 611885, chr8q24.3 (GRCh38 chr8:144,098,633–144,108,124, minus strand). MANE transcript NM_030974.4 / NP_112236.3 encodes a 387-aa scaffold protein.

Pathogenic variants (two reported, both homozygous frameshift LoF):

Patient	cDNA	Protein	Consequence	CADD	gnomAD
P1	c.220dupC	p.Leu74ProfsX86	Frameshift, truncation before/within UBL; NZF deleted	32.5	Absent
P2	c.613_614delCT	p.Leu205GlufsX21	Frameshift, truncation within UBL; NZF deleted	24.2	Absent

Both exceed the SHARPIN mutation-significance cutoff (MSC 12.1) and both produce **undetectable SHARPIN protein** (complete loss of function). No homozygous SHARPIN LoF or homozygous CNV loss is reported in **gnomAD**, indicating intolerance to complete loss. ACMG classification: pathogenic (null variant in a gene with established LoF mechanism; absent from population databases; segregating with disease in consanguineous pedigrees).

Variant type/class. Frameshift/truncating (null). **Origin:** germline. **Functional consequence:** loss of function (destabilized/absent LUBAC accessory subunit).

Protein domain architecture (why truncation abolishes function). SHARPIN (387 aa) has three domains: - **N-terminal PH (pleckstrin homology)** — integrin/tumor regulation; LUBAC-independent; not ubiquitin-binding. - **Central UBL (ubiquitin-like)** — binds the HOIP UBA/NZF2 region to assemble and stabilize LUBAC and relieve HOIP autoinhibition. - **C-terminal NZF (Npl4 zinc finger)** — binds K63-linked and linear ubiquitin; mediates cell-death protection.

Both UBL and NZF are required to stimulate linear ubiquitylation (UBL I272A and NZF T358L/F359V mutants abolish activity). The disease alleles p.Leu74Profs*86 and p.Leu205Glufs*21 truncate before/within the UBL and delete the NZF — abolishing LUBAC-supporting function and yielding undetectable protein.

Modifier genes. None established in humans. In mouse, death-pathway genes (*Tnfr1/Tnfrsf1a*, *Ripk3*, *Casp8*, *Mlkl*, *Tlr3*) genetically modify the phenotype (see Sections 6 and 15).

Epigenetic / chromosomal abnormalities. None implicated; AIFID is a single-gene LoF disease, not a chromosomal disorder.

Section 5 — Environmental Information

No environmental toxins, radiation, pollution, occupational exposures, or lifestyle factors are established as causing or triggering AIFID. No infectious agent causes the disease. The disease is a cell-intrinsic consequence of SHARPIN loss; TNF-superfamily ligands (endogenous) are the proximal drivers of the inflammatory and cell-death phenotype. In the mouse model, **dsRNA in skin acting through TLR3** is an environmental-like trigger for keratinocyte death and dermatitis (PMID: 27810922), but this branch is not evident in human disease.

Section 6 — Mechanism / Pathophysiology

Ordered causal chain

1. **Biallelic SHARPIN LoF variants** (frameshift, null) → **absent SHARPIN protein** (demonstrated: undetectable protein).
2. Absent SHARPIN → **destabilization and reduced abundance of LUBAC** (HOIP/HOIL-1L/SHARPIN complex) (demonstrated in mouse MEFs/B cells; inferred in human cells).

3. Reduced LUBAC → **diminished Met1 (M1)-linear polyubiquitination of NEMO (IKBKG) and RIPK1** (demonstrated biochemically for LUBAC generally).
4. Reduced linear ubiquitination of NEMO → **attenuated assembly/activation of the IKK complex → attenuated canonical NF-κB signaling** (demonstrated: patient fibroblasts and B cells show attenuated canonical NF-κB responses).
5. **Branch A (survival → death switch):** Attenuated NF-κB survival program + loss of LUBAC-mediated restraint on the TNFR1 complex → **increased propensity for TNF-superfamily-mediated cell death** via FADD/caspase-8-dependent apoptosis and RIPK3/MLKL-dependent necroptosis (demonstrated: patient cells show death propensity; mouse epistasis defines both death arms).
6. **Branch A → tissue injury:** Excess death of parenchymal/immune cells → **sterile multi-organ autoinflammation** (fever, arthritis, parotitis, colitis, hepatic glycogenosis). TNF is the key effector (demonstrated: anti-TNF produces complete resolution in P1; *Tnf* deletion prevents mouse skin lesions).
7. **Branch B (immunodeficiency):** Increased B-cell death → **reduced germinal-center B-cell development → humoral immunodeficiency** (demonstrated: both SHARPIN- and HOIP-deficient individuals show substantial GC B-cell reduction).
8. **Mouse-specific branch (not seen in humans):** Loss of LUBAC restraint on TLR3 → enhanced dsRNA-triggered keratinocyte death → **dermatitis** (demonstrated in *cpdm* mouse; absent in humans).

Molecular pathways

- **Canonical NF-κB pathway** (attenuated) — Reactome/KEGG NF-κB signaling; linear ubiquitination of NEMO is the proximal activating event.
- **TNFR1 signaling complex (Complex I/II)** — loss of LUBAC restraint tips the balance toward Complex II (death).
- **Apoptosis (FADD/caspase-8)** and **necroptosis (RIPK1/RIPK3/MLKL)** — the two death arms.

Cellular processes

Inflammation (GO:0006954), apoptotic process (GO:0006915), necroptotic process (GO:0070266), canonical NF-κB signal transduction (GO:0007249), linear/M1 protein polyubiquitination, B-cell differentiation / germinal-center formation (GO:0002467).

Protein dysfunction

Complete loss of the SHARPIN scaffold → LUBAC destabilization → loss of linear-ubiquitin E3 ligase activity support (HOIP is catalytic but requires SHARPIN/HOIL-1 for full activity and stability).

Immune system involvement

A dual lesion: **autoinflammation** (dysregulated innate TNF-driven inflammation and cell death) plus **immunodeficiency** (defective GC B cells, humoral deficiency). Notably, unlike HOIL-1/HOIP deficiency, SHARPIN-deficient monocytes do **not** show cytokine hyperresponsiveness.

Metabolic changes

Hepatic glycogenosis (abnormal glycogen accumulation in liver, by biopsy) — a feature paralleling other LUBAC defects (HOIP/HOIL-1 cause amylopectinosis/glycogen storage), linking M1-ubiquitin dysfunction to a glycogen-metabolism phenotype (mechanism not fully defined).

GO / CL term suggestions

- Biological process: GO:0007249 (I-κB kinase/NF-κB signaling), GO:0006915 (apoptotic process), GO:0070266 (necroptotic process), GO:0002467 (germinal center formation).
- Cell types: CL:0000236 (B cell), CL:0000844 (germinal center B cell), CL:0000057 (fibroblast), CL:0000235 (macrophage), CL:0000312 (keratinocyte — mouse branch).

Section 7 — Anatomical Structures Affected

Organ level (primary). Joints (sterile arthritis — ankle, atlanto-axial), salivary/parotid glands (parotitis), colon/gut (colitis), liver (glycogenosis), secondary lymphoid organs (adenoid/germinal centers). **Body systems:** musculoskeletal, digestive, hepatobiliary, immune/lymphoreticular.

Tissue/cell level. Germinal-center B cells (reduced; increased death), B cells and fibroblasts (attenuated NF-κB; propensity to die), macrophages/monocytes (involved but not hyperresponsive). In the mouse: keratinocytes (epidermal death → dermatitis).

Subcellular level. Cytoplasmic signaling platforms — the TNFR1 signaling complex and IKK complex at the plasma membrane/cytosol; LUBAC acts on NEMO in the cytosol. GO cellular component: GO:0005829 (cytosol), GO:0098797 (plasma membrane protein complex), LUBAC complex.

UBERON suggestions. UBERON:0001484 (parotid gland), UBERON:0000059 (large intestine/colon), UBERON:0002107 (liver), UBERON:0001474 (bone element/joint), UBERON:0001959 (adenoid/lymphoid tissue), UBERON:0002106 (spleen — mouse). **Lateralization:** arthritis can be bilateral; parotid involvement may be bilateral — not systematically described given small n.

Section 8 — Temporal Development

Onset. Childhood; parotid swelling noted from **age 5** in P1. Onset pattern is **episodic/recurrent** (fever flares) superimposed on chronic inflammation.

Progression. Untreated disease is progressive in disability (P1 wheelchair-dependent before treatment). Course pattern is **episodic/relapsing autoinflammation**. Duration is **chronic/lifelong** (genetic). Severity ranges from treatable (P1) to **fatal** (P2, death attributed to broader LUBAC-deficiency immunodeficiency despite anti-TNF reducing febrile episodes).

Patterns. Remission is **treatment-induced** by anti-TNF (sustained ~7-year remission in P1); no spontaneous remission documented. **Critical period for intervention:** early initiation of TNF blockade appears to control the autoinflammatory component, though it does not fully rescue the immunodeficiency.

Section 9 — Inheritance and Population

Epidemiology. Ultra-rare: **two patients reported worldwide** (PMID: 38609546). No meaningful prevalence/incidence estimate; effectively <1 per 10^7 (below Orphanet reporting thresholds).

Inheritance. Autosomal recessive — both patients homozygous, at least one in a consanguineous pedigree (Indian origin). **Penetrance:** presumed complete for biallelic null (both reported biallelic-null individuals are affected). **Expressivity:** variable (controllable vs

fatal). **Carrier frequency:** heterozygous SHARPIN LoF is tolerated (heterozygous carriers unaffected); no homozygous LoF in gnomAD. **Founder effect / anticipation / mosaicism:** none described (n too small). **Consanguinity** is relevant to recurrence risk.

Population demographics. One patient of **Indian** ancestry; no ethnic predilection can be inferred from n=2. **Sex ratio:** at least P1 was male (a boy) — not interpretable as a sex bias given sample size. **Age distribution:** pediatric onset.

Section 10 — Diagnostics

Clinical/laboratory tests. - Inflammatory markers during febrile flares (elevated acute-phase response — expected for autoinflammation). - **Sterile synovial fluid** on joint aspiration (arthritis is non-infectious). - **Liver biopsy** showing glycogenosis-like changes. - Immunophenotyping: reduced germinal-center B cells (adenoid/lymphoid tissue histology shows hypoplastic GCs); humoral immunodeficiency workup. - Functional cellular assays: **attenuated canonical NF-κB response** in patient fibroblasts and B cells; **increased TNF-superfamily-induced cell death** in patient cells. Absence of monocyte cytokine hyperresponsiveness helps distinguish from HOIL-1/HOIP deficiency. - **Western blot:** undetectable SHARPIN protein; reduced LUBAC.

Genetic testing (definitive). WES or WGS identifies the biallelic SHARPIN frameshift variants; inborn-errors-of-immunity / autoinflammatory **gene panels** that include SHARPIN, RNF31 (HOIP), RBCK1 (HOIL-1), OTULIN, TNFAIP3, and IKBKG are appropriate. Single-gene SHARPIN testing is reasonable when LUBAC disease is strongly suspected. CMA/karyotype/FISH/mtDNA/repeat testing are **not indicated** (point/frameshift LoF disease).

Omics-based support. Transcriptomic profiling was used to document resolution of the inflammatory signature on anti-TNF therapy and is informative for monitoring.

Clinical criteria / differential diagnosis. No formal diagnostic criteria exist (ultra-rare). Differential diagnosis spans the **LUBAC/M1-ubiquitin disease family and related NF-κB disorders** (see Section 15 table): HOIP (RNF31) deficiency, HOIL-1 (RBCK1) deficiency, OTULIN-related autoinflammatory syndrome (ORAS/otulipenia), A20/TNFAIP3 haploinsufficiency (HA20), and NEMO/IKBKG defects. **Discriminating features of AIFID:** autoinflammatory arthritis/parotitis/colitis + hepatic glycogenosis + GC B-cell immunodeficiency, with **no dermatitis** and **no monocyte cytokine hyperresponsiveness**.

Screening. Cascade genetic testing of relatives and carrier testing in consanguineous families; no newborn-screening program exists.

Section 11 — Outcome / Prognosis

Survival/mortality. Variable and component-dependent: P1 achieved **sustained remission (~7 years symptom-free)** on anti-TNF; P2 had a **fatal outcome** despite anti-TNF (death attributed to the immunodeficiency/broader LUBAC-deficiency features rather than uncontrolled fever). No formal survival statistics (n=2).

Morbidity/function. High pre-treatment morbidity (wheelchair dependence from arthritis in P1), reversible with TNF blockade for the autoinflammatory component.

Disease course/complications. Autoinflammation (controllable), humoral immunodeficiency (persists; predisposes to infection), and the metabolic (hepatic glycogenosis) feature. The immunodeficiency is the principal residual risk even when inflammation is controlled.

Prediction. Response to anti-TNF predicts control of autoinflammation; severity of the immunodeficiency component appears to drive the worst outcomes. **No validated prognostic biomarkers** beyond the inflammatory transcriptomic signature.

Section 12 — Treatment

Pharmacotherapy — mechanism-based. TNF inhibitors (anti-TNF) are the cornerstone. One patient achieved **complete clinical and transcriptomic resolution of autoinflammation** on anti-TNF; etanercept maintained ~7-year remission in P1. This is directly predicted by the mechanism (TNF-superfamily-driven cell death/inflammation downstream of LUBAC loss) and corroborated by the mouse model, where *Tnf* deletion prevents disease. - NCIT suggestions: Etanercept (NCIt C1119), TNF inhibitor / Tumor Necrosis Factor Inhibitor therapy, anti-TNF monoclonal antibodies (infliximab, adalimumab) as alternatives.

Pharmacogenomics. Not applicable in a conventional sense; treatment selection is driven by the genotype-defined mechanism (genotype-guided use of anti-TNF).

Advanced therapeutics. None approved specifically. Conceptually, **hematopoietic stem cell transplantation** could address the immunodeficiency component (as considered for other LUBAC defects), and gene correction is a theoretical future option — neither is established for AIFID. No RNA, targeted, or immunotherapy agents are specific to this disease.

Surgical/supportive/rehabilitative. Symptomatic management of arthritis (physical therapy, mobility support pre-treatment); immunoglobulin replacement and infection prophylaxis may be indicated for the humoral immunodeficiency.

Experimental / trials. No disease-specific registered trials (ultra-rare). Management is extrapolated from the broader autoinflammatory and inborn-errors-of-immunity fields.

Treatment strategy. (1) Confirm genetic diagnosis; (2) initiate **anti-TNF** to control autoinflammation; (3) manage the **immunodeficiency** (infection surveillance, immunoglobulin as needed); (4) monitor hepatic and inflammatory status. Anti-TNF controls inflammation but does **not** correct the underlying immunodeficiency — a key limitation illustrated by P2's fatal course.

Section 13 — Prevention

Primary prevention. Not applicable beyond **genetic counseling** in at-risk (consanguineous) families — carrier identification and reproductive options (prenatal/preimplantation genetic testing) can prevent recurrence.

Secondary/tertiary prevention. Early genetic diagnosis and prompt anti-TNF initiation prevent progression of disability (tertiary prevention of autoinflammatory complications). Infection prophylaxis mitigates immunodeficiency complications.

Immunization/public-health/environmental measures. Standard immunization caution applies to immunodeficient patients (live-vaccine considerations); no vaccine prevents the disease. No environmental interventions are relevant.

Counseling. Genetic counseling for autosomal-recessive recurrence (25% risk per pregnancy for carrier couples); cascade testing of relatives.

Section 14 — Other Species / Natural Disease

Taxonomy / orthologs. Mouse *Sharpin* (MGI:1913331; Entrez 109322; chromosome 15) is the principal ortholog studied (NCBI Taxon 10090, *Mus musculus*). Human *SHARPIN* NCBI Gene 81858 (Taxon 9606, *Homo sapiens*).

Natural disease in animals. The spontaneous mouse mutation **cpdm** (chronic proliferative dermatitis) is a naturally occurring *Sharpin*-null disorder — the founding model of SHARPIN biology — rather than a companion-animal/wildlife disease. No naturally occurring SHARPIN disease is documented in dogs, cats, or livestock in the reviewed literature.

Comparative biology. The core mechanism (LUBAC → M1-ubiquitin → NF-κB/cell-death control) is **evolutionarily conserved** between mouse and human. Comparative pathology shows **partial recapitulation with a key divergence**: mice develop prominent **dermatitis** (via TLR3/keratinocyte death) that humans lack, while both show multi-organ inflammation and lymphoid/immune defects.

Transmission. Not applicable (non-communicable genetic disease; no zoonotic potential).

Section 15 — Model Organisms

Principal model: the Sharpin-null mouse (cpdm). The spontaneous *Sharpin^cpdm/cpdm^* mutation causes **chronic proliferative dermatitis**: chronic eosinophilic dermatitis, multi-organ inflammation, splenomegaly, **loss of Peyer's patches / disrupted secondary lymphoid tissue**, and immunodeficiency with increased serum IgM (PMID: 21455180, PMID: 21455173).

Genetic epistasis defining the death pathways.

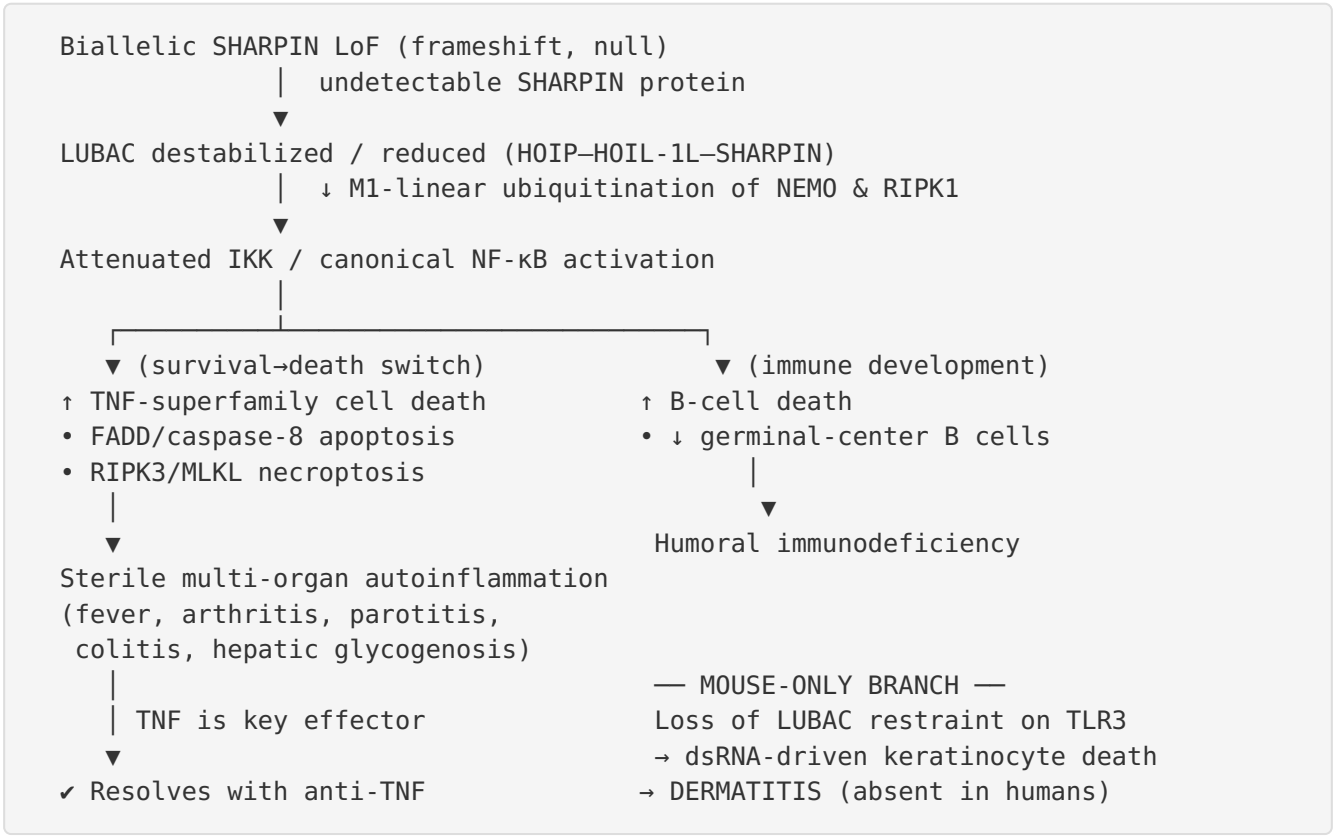
Genetic manipulation (on Sharpin-null background)	Effect on phenotype	PMID
Tnf deletion	Prevents skin lesions	21455173
Tnfr1 (not *Tnfr2*) deficiency	Suppresses multi-organ phenotype	25443632
Ripk3 or *Mlkl* deficiency	Partial amelioration (necroptosis arm)	25443632
Ripk3 deletion + *Casp8* heterozygosity	Near-complete suppression; restores Peyer's patches	25443632
Tlr3 ablation	Ameliorates dermatitis (keratinocyte death arm)	27810922
Integrin β1 function-blocking antibody	Reduces epidermal hyperproliferation	29040328

Model characteristics. The *cpdm* mouse **partially recapitulates** human disease: shared multi-organ inflammation, lymphoid/immune defects, attenuated NF-κB, and TNF-dependence (basis for anti-TNF therapy). **Key limitation:** the mouse's prominent **dermatitis is absent in humans**, driven by a TLR3/keratinocyte-death branch apparently not operative in human disease. The mouse also reveals a LUBAC-independent **integrin-inhibitory** role of SHARPIN contributing to epidermal hyperproliferation, and a LUBAC-independent **caspase-1-inhibitory** role relevant to sepsis (PMID: 26968342).

Available genetic models. Spontaneous *cpdm* allele; targeted *Sharpin* knockouts; compound crosses (*Tnfr1*$^{-/-}$, *Ripk3*$^{-/-}$, *Casp8*^+/−^, *Tlr3*$^{-/-}$, *Mlkl*$^{-/-}$). **Resources:** MGI (MGI:1913331), IMSR.

Cellular/in vitro models. Patient-derived fibroblasts and B cells (attenuated NF-κB, death propensity); HOIL-1-deficient patient fibroblasts used in related LUBAC studies; MEFs and B cells from *cpdm* mice.

Mechanistic Model / Interpretation



The unifying concept is that **SHARPIN loss converts a cell's default NF-κB-driven survival response to TNF into a death response**, simultaneously producing autoinflammation (from excess sterile cell death and TNF signaling) and immunodeficiency (from death of developing B cells). The therapeutic corollary — that **removing TNF collapses the pathogenic branch** — is validated both in the mouse (*Tnf* deletion prevents disease) and in the human patient (anti-TNF produces complete resolution). The human–mouse divergence (no human dermatitis) localizes to the TLR3/keratinocyte branch and is a cautionary note about over-extrapolating the *cpdm* model to human skin.

Evidence Base

PMID	Title (abbrev.)	Role in this report
38609546	*Biallelic human SHARPIN loss of function induces autoinflammation and immunodeficiency*	**Defining human study** (2 patients): disease definition, phenotype, variants, attenuated NF-κB, death propensity, GC B-cell defect, anti-TNF response, no dermatitis
25443632	*TNFR1-dependent cell death drives inflammation in Sharpin-deficient mice*	Mouse epistasis: TNFR1-dependence; apoptosis + necroptosis arms
21455173	*Linear ubiquitination prevents inflammation and regulates immune signalling*	cpdm phenotype; *Tnf* deletion prevents skin lesions
21455181	*SHARPIN forms a linear ubiquitin ligase complex regulating NF-κB and apoptosis*	SHARPIN as 3rd LUBAC subunit; NEMO linear ubiquitination; FADD/caspase-8 death
21455180	*SHARPIN is a component of the NF-κB-activating LUBAC*	SHARPIN loss reduces LUBAC; attenuates NF-κB in MEFs/B cells
22863777	*HOIP specifies linear ubiquitin chain assembly (RBR + LDD)*	HOIP is catalytic; SHARPIN/HOIL-1 accessory
24469399	*Mechanism of IKK activation by LUBAC*	Linear-ubiquitinated NEMO activates IKK/NF-κB
23969028	*Linear ubiquitination-mediated NF-κB regulation and disorders*	Counter-regulatory DUBs CYLD, A20, OTULIN
27810922	*LUBAC deficiency perturbs TLR3 signaling*	Mouse-specific TLR3/keratinocyte-death dermatitis branch
26008899	*Human HOIP/LUBAC deficiency*	Differential diagnosis: HOIP deficiency
30804083	*OTULIN deficiency in ORAS*	Differential diagnosis: ORAS (M1-ubiquitin family)
29040328	*Integrin β1 inhibition alleviates cpdm dermatitis*	LUBAC-independent integrin role of SHARPIN
26968342	*SHARPIN regulates caspase-1 in sepsis*	LUBAC-independent SHARPIN function (caspase-1 inhibition)

Verified direct quotes supporting key claims: - Disease definition — [PMID: 38609546]: *"Here, we report two individuals with SHARPIN deficiency who manifest autoinflammatory symptoms but unexpectedly no dermatological problems. Fibroblasts and B cells from these individuals showed attenuated canonical NF-κB responses and a propensity for cell death mediated by TNF superfamily members."* - Therapy — [PMID: 38609546]: *"Treatment of one SHARPIN-deficient individual with anti-TNF therapies led to complete clinical and transcriptomic resolution of autoinflammation."* - Immunodeficiency — [PMID: 38609546]: *"Both SHARPIN-deficient and HOIP-deficient individuals showed a substantial reduction of secondary lymphoid germinal center B cell development."* - Mouse model — [PMID: 25443632]: *"The inactivating mouse Sharpin cpdm mutation causes TNF-dependent multi-organ inflammation, characterized by dermatitis, liver inflammation, splenomegaly, and loss of Peyer's patches."* - Death pathways — [PMID: 25443632]: *"Ripk3 or Mlkl deficiency partially ameliorated the multi-organ phenotype, and combined Ripk3 deletion and Casp8 heterozygosity almost completely suppressed it, even restoring Peyer's patches."* - Catalytic subunit — [PMID: 22863777]: *"both LUBAC catalytic activity and LUBAC specificity for linear ubiquitin chain formation are embedded within the RING-IBR-RING (RBR) ubiquitin ligase subunit HOIP."* - NEMO/IKK — [PMID: 24469399]: *"linearly polyubiquitinated NEMO activates IKK more potently than unanchored linear chains."* - Mouse NF-κB — [PMID: 21455180]: *"Deletion of SHARPIN drastically reduced the amount of LUBAC, which resulted in attenuated TNF-α- and CD40-mediated activation of NF-κB in mouse embryonic fibroblasts (MEFs) or B cells from cpdm mice."*

Limitations and Knowledge Gaps

1. **Extremely small human evidence base (n=2).** All human-specific conclusions rest on a single 2024 study of two unrelated patients. Prevalence, penetrance, expressivity, sex bias, age distribution, and the full phenotypic spectrum cannot be reliably estimated.
2. **Human–mouse divergence.** The principal model (*cpdm* mouse) displays dermatitis that humans lack, so the mouse overstates skin involvement and the human disease's full organ spectrum may still be incompletely mapped.
3. **Mechanism of hepatic glycogenosis** downstream of LUBAC loss is unexplained (parallels HOIP/HOIL-1 amylopectinosis but mechanistically undefined).
4. **Immunodeficiency not corrected by anti-TNF.** P2's fatal course indicates anti-TNF controls inflammation but not the immunodeficiency; optimal management of the immune-deficiency component (e.g., HSCT) is undefined.

5. **No modifier genes, no validated biomarkers** in humans; prognostic stratification is anecdotal.
6. **No registered clinical trials** or approved disease-specific therapeutics beyond repurposed anti-TNF.

Proposed Follow-up Experiments / Actions

1. **Case ascertainment / international registry.** Use GeneMatcher and inborn-errors-of-immunity networks (IUIS) to find additional SHARPIN-deficient patients and build a natural-history dataset; this is the single highest-value action.
2. **Define the immunodeficiency mechanism.** Deep immunophenotyping of germinal-center reactions and B-cell survival in patient and *Sharpin*-null tissues; test whether B-cell-intrinsic death (caspase-8/RIPK) drives the GC defect, and whether RIPK1 kinase inhibition rescues it.
3. **Dissect the human-vs-mouse dermatitis divergence.** Compare TLR3/dsRNA-induced keratinocyte death in human SHARPIN-deficient vs *cpdm* keratinocytes to explain why humans are spared.
4. **Test next-generation targeted therapies.** Evaluate RIPK1 kinase inhibitors and combined apoptosis/necroptosis blockade in patient cells and mouse models, given epistasis showing both death arms contribute — potentially addressing features not fully controlled by anti-TNF.
5. **Evaluate HSCT** as definitive therapy for the immunodeficiency component in severe cases, informed by experience in HOIP/HOIL-1 deficiency.
6. **Investigate the glycogenosis phenotype** metabolically (liver glycogen structure/quantity) to connect M1-ubiquitin loss to glycogen-storage pathology across the LUBAC disease family.

Consensus Answer

Autoinflammation with episodic fever and immune dysregulation (AIFID; MONDO:0968982, OMIM 620795) is an ultra-rare autosomal-recessive inborn error of immunity caused by biallelic loss-of-function variants in **SHARPIN** (HGNC:25321, 8q24.3), the accessory subunit of the linear ubiquitin chain assembly complex (LUBAC). Loss of SHARPIN destabilizes LUBAC and reduces M1-linear ubiquitination of NEMO/RIPK1, attenuating canonical NF-κB signaling

and shifting cells toward TNF-superfamily-driven apoptosis/necroptosis — producing childhood-onset episodic fever, sterile arthritis, parotitis, colitis, hepatic glycogenosis, and germinal-center B-cell immunodeficiency (but, unlike the Sharpin-null *cpdm* mouse, no dermatitis), with the autoinflammation responding dramatically to TNF inhibition. The disease is defined by two patients reported by Oda et al. 2024 (PMID: 38609546).
